SMOKE DETECTOR SYSTEM

STEM PROJECT REPORT

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BONAFIDE CERTIFICATE :

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INTERNAL EXAMINER	**EXTERNAL EXAMINER**

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ABSTRACT:

Approximation methods exist to provide estimates of smoke detector response based on optical density, temperature rise, and gas velocity thresholds. The objective of this study was to assess the uncertainty associated with three estimation methods. Experimental data was used to evaluate recommended alarm thresholds and to quantify the associated error. With few exceptions less than 50 percent of the predicted alarm times occurred within 60 seconds of the experimental alarms. At best errors of 20 to 60 percent (in underprediction) occurred for soldering fires using an optical density threshold. For flaming fires, errors in predicted alarm times on the order of 100 to 1000 percent in over prediction of the experimental alarms were common. Overall, none of the approximation methods distinguished themselves as vastly supervisor. Great care must be exercised when applying these approximation methods to ensure that uncertainty in the predicted alarm times is appropriately considered. The advanced smoke detector system detailed in this study represents a significant leap forward in fire safety technology. Utilizing a combination of photoelectric and ionization sensors, the system ensures the early detection of a wide range of smoke particles, providing comprehensive coverage for various fire types. Once smoke is detected, the system activates an audible alarm and sends real-time notifications to designated smartphones through a dedicated mobile application. This application offers users the ability to remotely monitor their environment, perform system diagnostics, and receive maintenance reminders. The system's mesh network capability allows multiple detectors to communicate, creating an extensive and reliable safety net that ensures any triggered alarm activates all connected detectors. Additionally, the system includes environmental sensors to monitor temperature, humidity, and air quality, enhancing detection accuracy and minimizing false alarms. By integrating IoT technology, the system connects seamlessly with smart home systems and central monitoring services, enabling automated emergency responses such as unlocking doors, shutting down HVAC systems, and alerting fire departments. Field tests have demonstrated the system's high sensitivity, reliability, and ease of use, highlighting its potential to significantly improve fire safety standards and response times, thus reducing the risk of fire-related injuries and property damage.

CHAPTER 1

INTRODUCTION ABOUT THIS PROJECT

A **smoke detector** is a device that sense smoke, typically as an indicator of fire. Commercial security devices issue a signal to a fire alarm control panel as a part of a fire alarm system, while household smoke detectors, also known as smoke **alarms**, generally issue a local audible or visual alarm from the detector itself or several detectors if there are multiple smoke detectors interlinked. Smoke detectors are housed in plastic enclosures, smoke detectors are typically shaped like a disk or square about 150 millimeters (6 in) in diameter and 25 millimeters (1 in) thick, but shape and size vary. Smoke can be detected either optically (photoelectric) or by physical process (ionization): detectors may use either, or both, methods. Sensitive alarms can be used to detect, and thus deter, smoking in areas where it is banned. Smoke detectors in large commercial, industrial, and residential buildings are usually powered by a central fire alarm system, which is powered by the building power with a battery backup. Domestic smoke detectors range from individual battery-powered units, to several interlinked mains-powered units with battery backup; with these interlinked units, if any unit detects smoke, all trigger even if household power has gone out.

CHAPTER 2:

LITERATURE SURVEY:

Introduction

Smoke detector systems are crucial for early fire detection, providing timely warnings to prevent loss of life and property. This literature survey aims to explore the different types of smoke detectors, their underlying technologies, recent advancements, performance evaluation methods, standards, and applications. By reviewing existing research and technological developments, we seek to identify trends, gaps, and future directions in the field.

Types of Smoke Detectors :

Smoke detectors can be broadly categorized into ionization, photoelectric, and combination detectors. Ionization smoke detectors use a small amount of radioactive material to ionize air in a sensing chamber, making them highly sensitive to small smoke particles typically produced by flaming fires. Photoelectric

provide comprehensive coverage and enhance reliability in detecting various types of fires.

Detection Technologies

Advances in detection technologies have significantly improved the performance of smoke detectors. Optical sensing technologies have evolved, employing advanced algorithms and sensors to enhance accuracy and reduce false alarms. Ionization sensing has also seen improvements in sensitivity and energy efficiency. Aspirating smoke detection systems, which continuously sample air from the protected area, offer high sensitivity and are particularly useful in environments where early detection is critical. Multi-criteria detectors, which combine different sensing technologies such as heat, carbon monoxide, and smoke detection, provide robust solutions for complex environments.

Performance Evaluation

Evaluating the performance of smoke detectors involves examining sensitivity, specificity, response time, and environmental impact. Sensitivity refers to the detector's ability to identify smoke at low concentrations, while specificity measures its ability to distinguish between smoke and other airborne particles. Response time is critical in determining how quickly a detector can alert occupants to the presence of smoke. Additionally, the performance of smoke detectors in various environments, such as residential, commercial, and industrial settings, is crucial for ensuring their effectiveness under different conditions.

Standards and Regulations

Industry standards and regulations play a vital role in shaping the design and deployment of smoke detector systems. Key standards, such as the National Fire Protection Association (NFPA) 72 and Underwriters Laboratories (UL) 268, outline the performance criteria and testing methods for smoke detectors. These standards ensure that detectors meet minimum safety requirements and perform reliably in realworld conditions. Regulatory bodies mandate compliance with these standards, influencing the development and market availability of smoke detector technologies.

Applications :

Smoke detectors are used in a variety of settings, each with specific requirements. In residential applications, smoke detectors are designed for ease of installation and maintenance, with a focus on cost-effectiveness and reliability. Commercial and industrial applications often require more sophisticated systems with enhanced features such as networked alarms and integration with building management systems. Specialized environments, such as aircraft, ships, and hazardous locations, demand customized solutions that can withstand challenging conditions and provide early and accurate detection.

Conclusion :

The literature survey highlights significant advancements in smoke detector technologies and underscores the importance of ongoing research to address existing gaps and challenges. Improved sensitivity, specificity, and response times are critical areas for future development. Adhering to industry standards and regulations ensures the reliability and safety of smoke detectors. Future research should focus on enhancing multi-criteria detection systems, reducing false alarms, and developing cost-effective solutions for various applications. By advancing smoke detector technology, we can better protect lives and property from the devastating effects of fires.

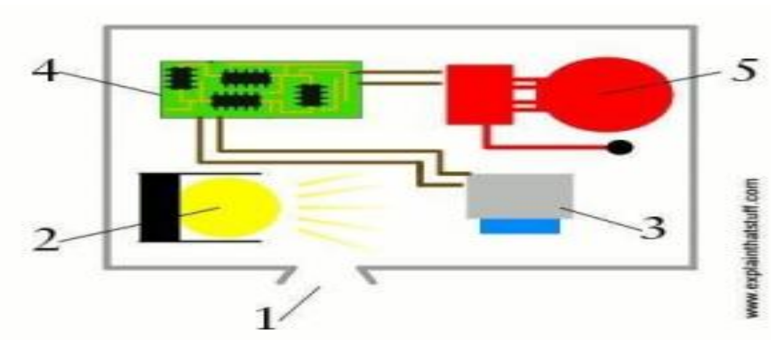
CHAPTER 3:

PRINCIPLE:

The detector must be screwed to your ceiling because that's where smoke heads for when something starts to burn. Fire generates hot gases and because these are less dense (thinner—or weigh less per unit of volume) than ordinary air they rise upward, swirling tiny smoke particles up too. As you can see in the photo up above, the detector has slits around its case (1), which lead to the main detection chamber. An invisible, infared light beam, similar to the ones that Tom Cruise dodged, shoots into the chamber from a **light-emitting diode (LED)** (2). The same chamber contains a photocell (3), which is an electronic light detector that generates electricity when light falls on it. Normally, when there is no smoke about, the light beam from the LED does not reach the detector. An electronic circuit (4), monitoring the photocell, detects that all is well, and nothing happens. The alarm (5) remains silent.

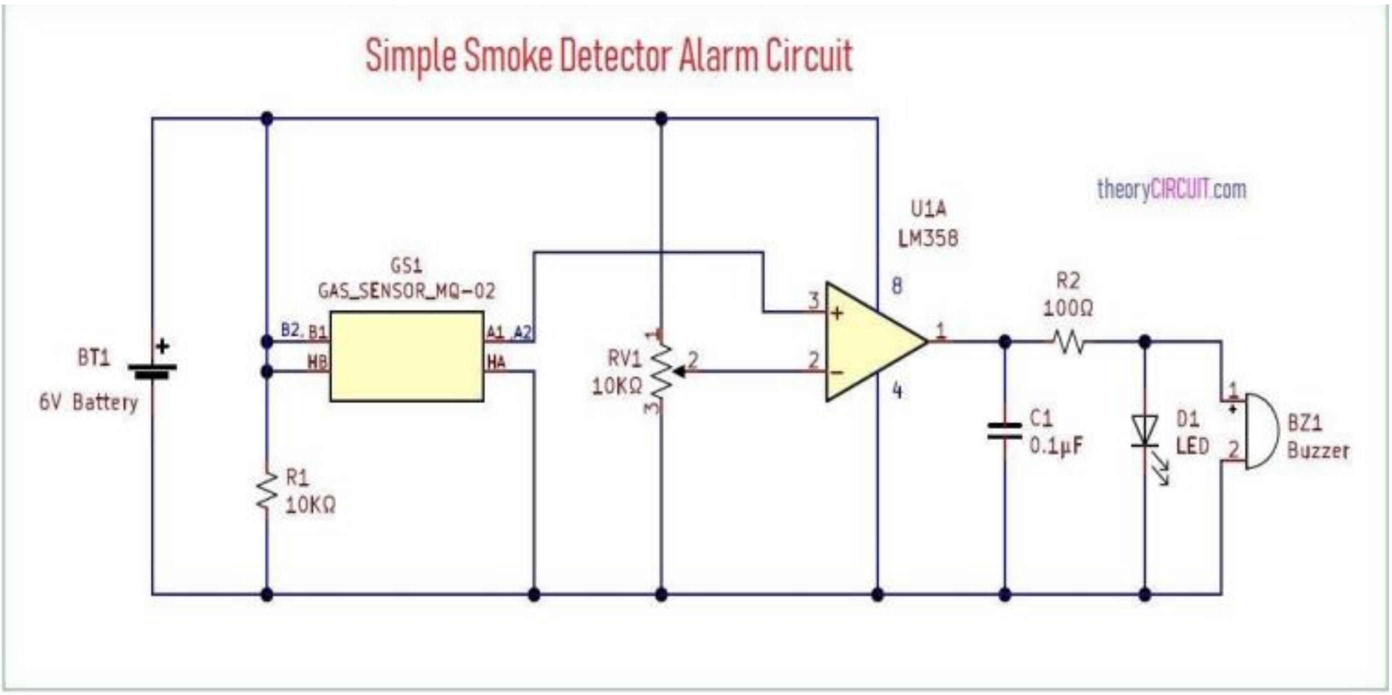
But if a fire breaks out, smoke enters the chamber (6) and scatters some of the light beam (7) into the photocell (3). This triggers the circuit (8), setting off the shrill and nasty alarm (9) that wakes you up and saves your life

BLOCK DIAGRAM:

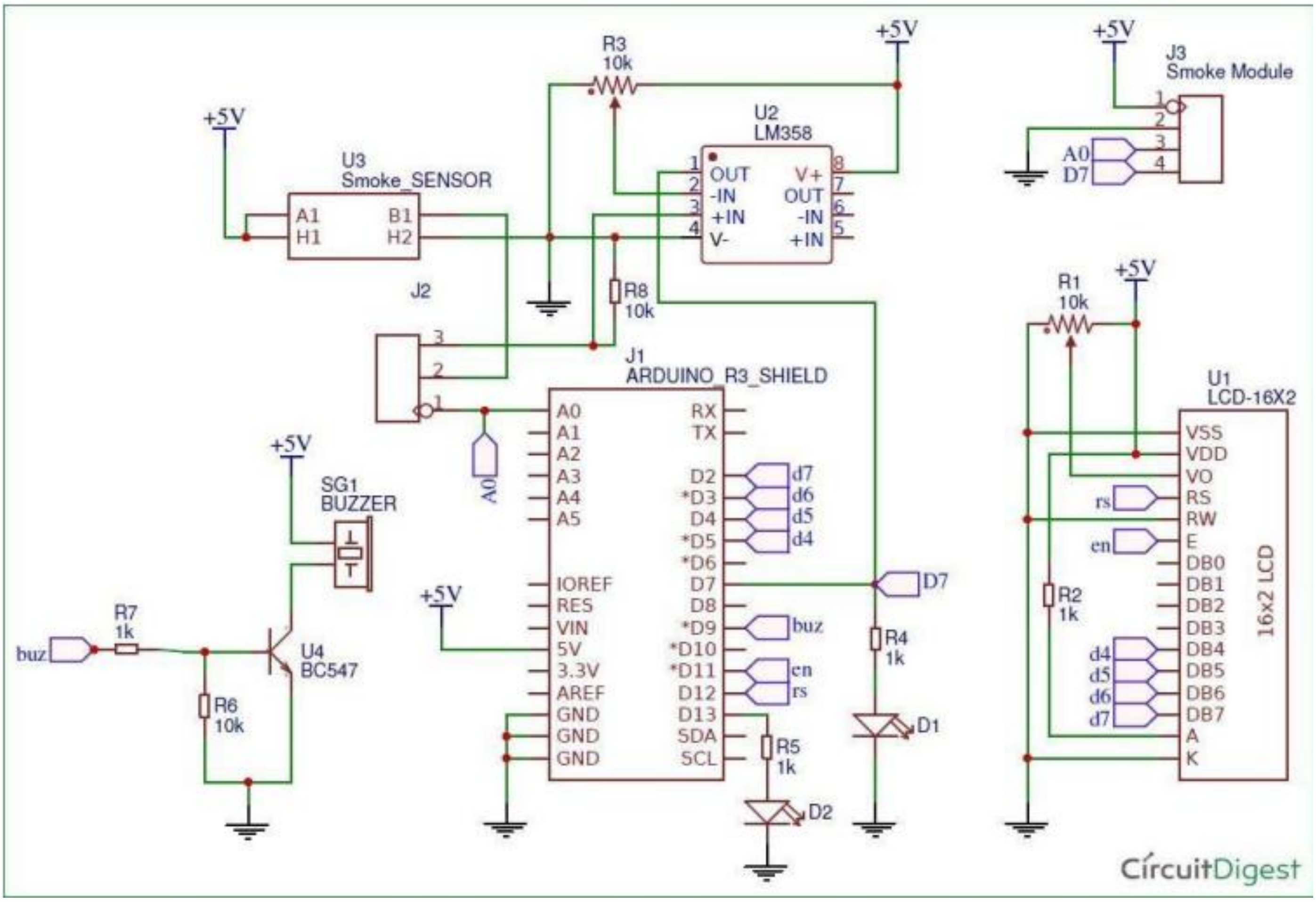


Simple Smoke Detector Alarm circuit using MQ02 designed with few easily available components. Main part of this circuit is Gas sensor MQ-02 which is capable of detecting Smoke, LPG, Propane and Hydrogen. It can be used in different types of applications where the smoke and other mentioned gas leak detection required.

The following prototype is a test circuit and it can be constructed through breakout board and PCB after calibration of each component. This smoke detector circuit will produce visible and audible alert when smoke detected.



CONSTRUCTION:-



In this Smoke Detector Circuit with Arduino, we have used a MQ2 Gas Sensor to detect preset smoke in the air. A 16x2 LCD is used for displaying the PPM value of Smoke. And an LM358 IC for converting smoke sensor output into digital form (this function is optional). A buzzer is placed as an alarm which gets triggered when smoke level goes beyond 1000 PPM. Circuit connections for this project are very simple, we have a Comparator Circuit for comparing output voltage of smoke sensor with preset voltage (output connected at pin D7). Also smoke sensor output is connected at an analog pin of Arduino (A0). Buzzer is connected at Pin D9. And LCD connections are same as Arduino LCD examples that are available in Arduino IDE (12, 11, 5, 4, 3, 2). Remaining connections are shown in the circuit diagram.

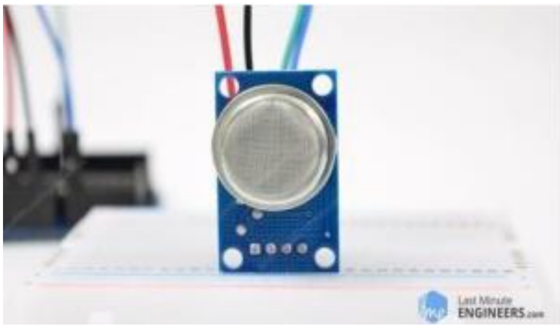
COMPONENTS REQUIRED:

- NODEMCU
- Bread Board
- MQ-2 Smoke Sensor
- Jumper Wires
- Red Led
- Green Led
- Resistor

NODEMCU:

Node MCU is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. In this crystal ball project arduino is used to display the message on the LCD according to the code written in the arduino software, Whenever the tilt switch senses the moment of bread board and sends the signals to the arduino.

MQ-2 SMOKE SENSOR:



Give your next Arduino project a nose for gasses with the MQ2 Gas Sensor Module. This is a robust Gas sensor suitable for sensing LPG, Smoke, Alcohol, Propane, Hydrogen, Methane and Carbon Mono oxide concentrations in the air. If you are planning on creating an indoor air quality monitoring system; breath checker or early fire detection system, MQ2 Gas Sensor Module is a great choice. MQ2 is one of the commonly used gas sensors in MQ sensor series. It is Metal Oxide Semiconductor (MOS) type Gas Sensor also known as semi resistors as the detection is based upon change of resistance of the sensing material when the Gas comes in contact with the material. Using a simple voltage divider network, concentrations of gas can be detected.

MQ-2 SENSOR

MQ2 Gas sensor works on 5V DC and draws around 800mW. It can detect LPG, Smoke, Alcohol, Propane, Hydrogen, Methane and Carbon Monoxi de concentrations anywhere from 200 to 10000ppm.

Here are the complete specifications

Operating Voltage	5v
Load Resistance	20 KΩ
Heater Resistance	33Ω ± 5%
Heating Consumption	<800mw
Sensing Resistance	10 KΩ – 60 KΩ
Concentration Scope	200 – 10000ppm
Preheat Time	Over 24 hours

CHAPTER 4 :

Existing system:

How They Work: These detectors use a small amount of radioactive material to ionize air within a detection chamber. This ionized air allows a small electric current to flow between two electrodes. When smoke enters the chamber, it disrupts the flow of ions, reducing the current and triggering the alarm. Advantages: They are particularly good at detecting fast-flaming fires. Disadvantages: They can be prone to false alarms from cooking smoke or steam and may not be as effective at detecting slow, smoldering fires.2. Photoelectric Smoke Detectors How They Work: These detectors use a light source and a light sensor. When smoke enters the chamber, it scatters the light, which is then detected by the sensor, triggering the alarm. Advantages: They are more responsive to slow, smoldering fires. Disadvantages: They might be less effective for fast-flaming fires compared to ionization detectors.3. Dual-Sensor Smoke Detectors How They Work: These combine both ionization and photoelectric technologies, aiming to offer broader detection capabilities for different types of fires. Advantages: They provide comprehensive coverage and are effective at detecting both fast-flaming and slow-smoldering fires. Disadvantages: They can be more expensive and still subject to the individual limitations of each sensor type.4. Aspirating Smoke Detectors (ASD)How They Work: These systems actively draw air through a network of pipes into a detection chamber where it is analyzed for the presence of smoke particles. Advantages: They are highly sensitive and can detect smoke at very low concentrations, making them ideal for high-value or high-risk areas. Disadvantages: They are more complex and expensive to install and maintain.5. Heat Detectors How They Work: These devices detect an increase in temperature caused by a fire. There are two main types: fixed temperature and rate-of-rise heat detectors. Advantages: They are less prone to false alarms from non-fire-related smoke (like cooking smoke). Disadvantages: They do not detect smoke and therefore are slower to respond to smoldering fires.6. Combination Smoke and Carbon Monoxide Detectors How They Work: These devices integrate both smoke and carbon monoxide detection into a single unit, providing dual protection. Advantages: They save space and can be more cost-effective than installing separate detectors. Disadvantages: They must be installed in locations where both smoke and carbon monoxide are likely to be detected effectively.7. Smart Smoke Detectors How They Work: These detectors can connect to a home's Wi-Fi network, allowing for remote monitoring and control via smartphone apps. They often include features like voice alerts, self-testing capabilities, and integration with other smart home devices. Advantages: Enhanced features and convenience, ability to receive alerts even when away from home. Disadvantages: Higher cost and dependency on reliable internet connectivity. Key Features Across Systems: Power Source:

Smoke detectors can be battery-operated, hardwired, or a combination of both with battery backup. Interconnectivity: Some systems allow detectors to be interconnected so that when one detector senses smoke, all alarms in the network are triggered. Sensitivity Adjustments: Advanced models may allow sensitivity adjustments to reduce false alarms while maintaining detection accuracy. Regulatory and Maintenance Considerations: Regulations: Local building codes and regulations often specify the type and placement of smoke detectors in residential and commercial buildings. Maintenance: Regular testing and maintenance are crucial to ensure smoke detectors function correctly. This includes periodic battery replacement, cleaning, and testing of alarm functionality. Smoke detector systems are critical components of fire safety, and choosing the right type depends on the specific needs of the environment they are protecting.

Internal structure of MQ2 Gas Sensor:

It also provides protection for the sensor and filters out suspended particles so that only gaseous elements are able to pass inside the chamber. The mesh is bound to rest of the body via a copper plated clamping ring

This is how the sensor looks like when outer mesh is removed. The star-shaped structure is formed by the sensing element and six connecting legs that extend beyond the Bakelite base. Out of six, two leads (H) are responsible for heating the sensing element and are connected through Nickel-Chromium coil, well known conductive alloy. The remaining four leads (A & B) responsible for output signals are connected using Platinum Wires. These wires are connected to the body of the sensing element and convey small changes in the current that passes through the sensing element.

The tubular sensing element is made up of Aluminum Oxide (AL2O3) based ceramic and has a coating of Tin Dioxide (SnO2). The Tin Dioxide is the most important material being sensitive towards combustible gases. However, the ceramic substrate merely increases heating efficiency and ensures the sensor area is heated to a working temperature constantly.

So, the Nickel-Chromium coil and Aluminum Oxide based ceramic forms a Heating System; while Platinum wires and coating of Tin Dioxide forms a Sensing System

How does a gas sensor work?

When tin dioxide (semiconductor particles) is heated in air at high temperature, oxygen is adsorbed on the surface. In clean air, donor electrons in tin dioxide are attracted toward oxygen which is adsorbed on the surface of the sensing material. This prevents electric current flow.

In the presence of reducing gases, the surface density of adsorbed oxygen decreases as it reacts with the reducing gases. Electrons are then released into the tin dioxide, allowing current to flow freely through the sensor

Hardware Overview – MQ2 Gas Sensor:

Module:

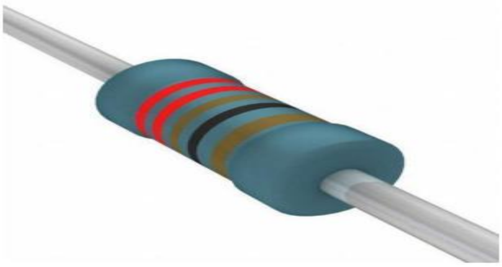
Since MQ2 Gas Sensor is not breadboard compatible, we do recommend this handy little breakout board. It's very easy to use and comes with two different outputs. It not only provides a binary indication of the presence of combustible gases but also an analog representation of their concentration in air.

The analog output voltage provided by the sensor changes in proportional to the concentration of smoke/gas. The greater the gas concentration, the higher is the output voltage; while lesser gas concentration results in low output voltage. The following animation illustrates the relationship between gas concentration and output voltage.

- VCC supplies power for the module. You can connect it to 5V output from your Arduino.
- GND is the Ground Pin and needs to be connected to GND pin on the Arduino.

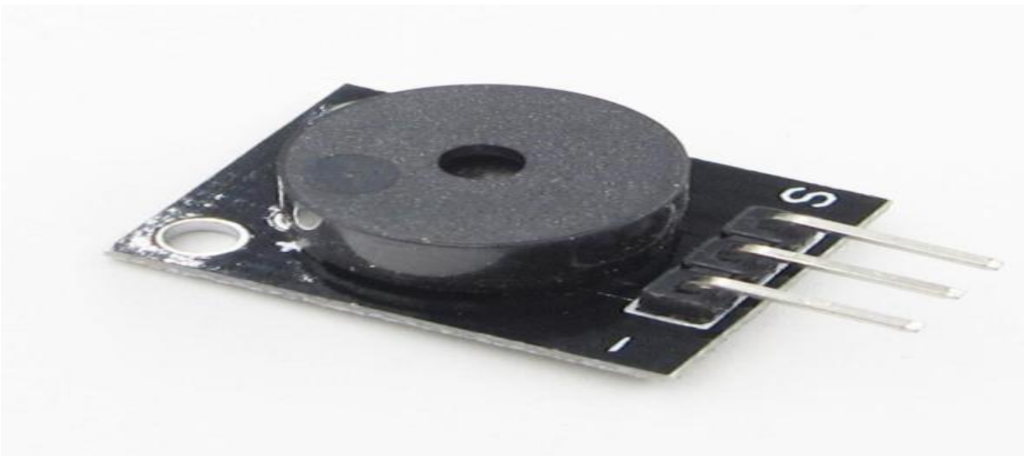
- D0 provides a digital representation of the presence of combustible gases
- A0 provides analog output voltage in proportional to the concentration of smoke/gas.

RESISTOR:



PASSIVE BUZZER:

A buzzer or beeper is an audio signalling device, which may be mechanical, electromechanical, or piezoelectric (piezo for short). Typical uses of buzzers and Beepers include alarm devices, timers, and confirmation of user input.



JUMPER WIRES:

- The term "jumper wire" simply refers to a conducting wire that establishes an electrical connection between two points in a circuit. You can use jumper wires to modify a circuit or to diagnose problems in a circuit. The following steps outline how you can safely use jumper wires in different electrical applications.

Connecting Wires:

- Connecting wires allows an electrical current to travel from one point on a circuit to another because electricity needs a medium through which it can move. Most of the connecting wires are made up of copper or aluminum. In this crystal ball project connecting wires are used to connect the circuit.

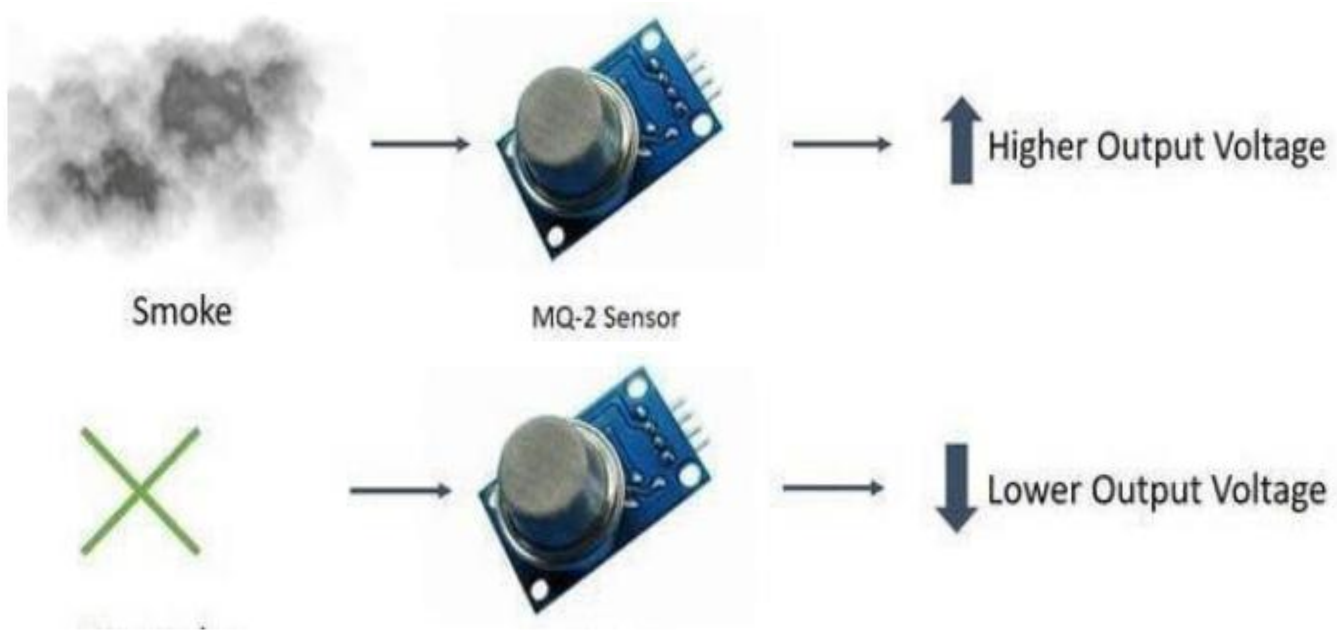
WORKING:

How does it Work?

- The voltage that the sensor outputs changes accordingly to the smoke/gas level that exists in the atmosphere. The sensor outputs a voltage that is proportional to the concentration of smoke/gas.
- In other words, the relationship between voltage and gas concentration is the following:
- The greater the gas concentration, the greater the output voltage

- The lower the gas concentration, the lower the output voltage

- The output can be an analog signal (A0) that can be read with an analog input of the Arduino or a digital output (D0) that can be read with a digital input of the Arduino.



Working Mechanism:

The MQ-2 sensor has 4 pins:

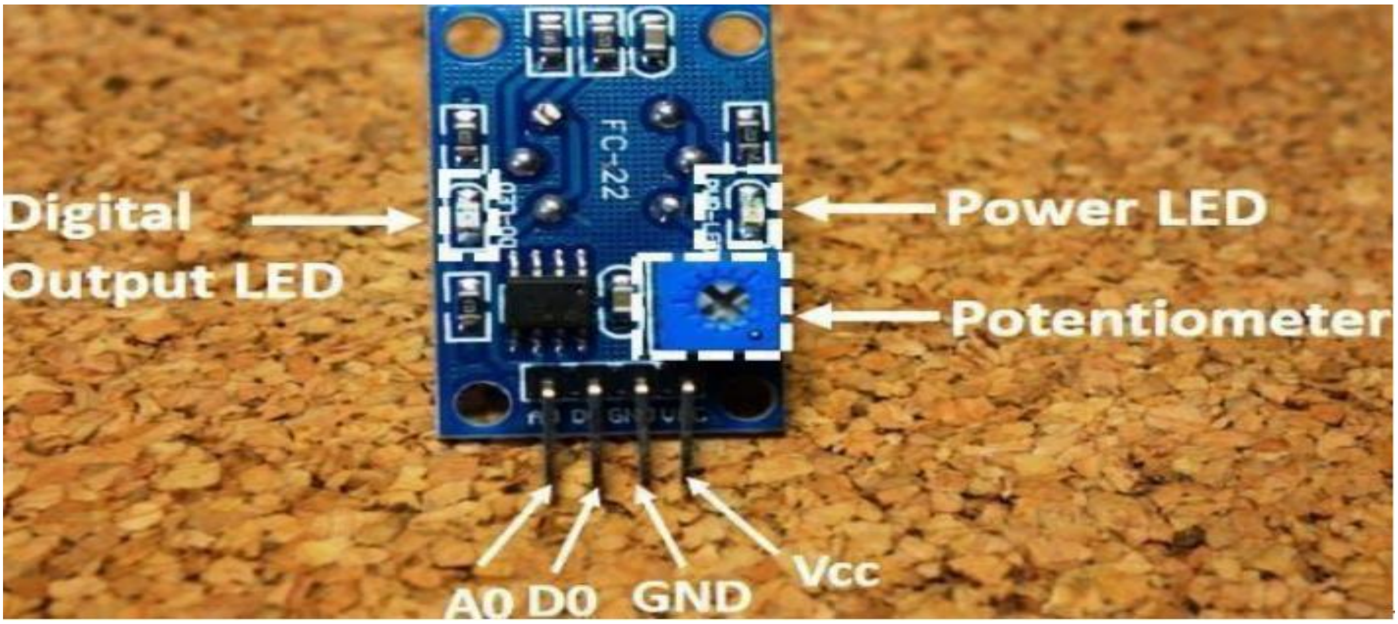
Pin--------------------------------------Wiring to Arduino Uno

A0--------------------------------------Analog pins

D0--------------------------------------Digital pins

GND-----------------------------------GND

VCC-----------------------------------5V



CODE:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// Define the I2C address for the LCD
LiquidCrystal_I2C lcd(0x27, 16, 2);  // Change 0x27 to the address found by the I2C scanner if different
```

```
// Define pin connections
const int mq2Pin = A0;         // Analog pin for MQ-2
const int buzzerPin = 8;       // Digital pin for Buzzer
const int ledContinuousPin = 6; // Digital pin for continuously glowing LED
const int ledBuzzerPin = 7;     // Digital pin for LED that turns on with buzzer
int gasThreshold = 300;         // Gas level threshold, adjust as needed

void setup() {
  // Initialize serial communication for debugging
  Serial.begin(9600);

  // Initialize pins
  pinMode(mq2Pin, INPUT);
  pinMode(buzzerPin, OUTPUT);
  pinMode(ledContinuousPin, OUTPUT);
  pinMode(ledBuzzerPin, OUTPUT);

  // Turn on the continuous LED
  digitalWrite(ledContinuousPin, HIGH);

  // Initialize the LCD
  lcd.begin(16, 2);  // Initialize the LCD with 16 columns and 2 rows
  lcd.backlight();   // Turn on the LCD backlight
  // Print a message to the LCD
  lcd.print("Initializing...");
}

void loop() {
  // Read the analog value from the MQ-2 sensor
  int gasLevel = analogRead(mq2Pin);

  // Print the gas level to the Serial Monitor
  Serial.print("Gas Level: ");
  Serial.println(gasLevel);
```

```
  // Set the cursor to column 0, line 0

  lcd.setCursor(0, 0);
  // Print the gas level to the LCD

  lcd.print("Gas Level: ");
  lcd.print(gasLevel);
  lcd.print("   "); // Clear any remaining characters

  // Check if gas level exceeds the threshold

  if (gasLevel > gasThreshold) {
    // Turn on the buzzer and the second LED
    digitalWrite(buzzerPin, HIGH);
    digitalWrite(ledBuzzerPin, HIGH);
    Serial.println("Gas leak detected! Buzzer and LED activated.");
    // Set the cursor to column 0, line 1
    lcd.setCursor(0, 1);
    // Print message to the LCD
    lcd.print("Smoke Detected!");
  } else {
    // Turn off the buzzer and the second LED
    digitalWrite(buzzerPin, LOW);
    digitalWrite(ledBuzzerPin, LOW);
    Serial.println("Gas level normal. Buzzer and LED deactivated.");
    // Clear the second line of the LCD
    lcd.setCursor(0, 1);
    lcd.print("smoke deduct"); // Clear the line
  }

  // Small delay before next reading
  delay(500);
}
```

ADVANTAGES: -

- User friendly: we can edit the code whenever we want to change tone of the button.
-
- Efficiency: it uses less power of only 5v.

- Portable: it is very small to carry wherever you go

- Future Enhancement: This technology could be further modified and more upgraded as per individual need and interest. We have discussed some basic ideas of this technology. And depending on innovative applications user can upgrade as per requirement.

- Detection of low energy fires

- Detection that is faster than heat detectors for most fires
  Are preferred in life safety applications

RESULT: -

Hence the experiment of SMOKE DETECTOR is done by using MQ-2 smoke detecting sensor successfully and it is is display.

CONCLUSION:

In conclusion, a smoke detector system is an indispensable component of fire safety in both residential and commercial settings. These systems play a critical role in providing early warning of smoke and potential fires, allowing occupants to take timely action to evacuate and contact emergency services. The effectiveness of smoke detectors hinges on several key factors: Early Detection and

Warning: Smoke detectors can detect the presence of smoke long before it becomes a visible threat. This early warning system is vital in preventing fires from escalating, thus reducing potential damage and saving lives. Technological Advancements: Modern smoke detector systems often include features such as interconnected alarms, which ensure that if one detector is triggered, all alarms in the building will sound. This is particularly useful in larger buildings where the source of the smoke might be far from some occupants. Additionally, integrating smoke detectors with smart home systems allows for real-time alerts to be sent to occupants' smartphones, ensuring that they are informed even when they are not on the premises. Ulti-functional Detectors: Many contemporary smoke detectors also include carbon monoxide (CO) detection. CO is a colorless, odorless gas that can be lethal, and having a combined smoke and CO detector enhances the overall safety by addressing multiple hazards with a single device. Maintenance and Reliability: The reliability of smoke detector systems depends significantly on regular maintenance. This includes routine testing of the alarms, timely replacement of batteries, and ensuring that detectors are free from dust and debris that could impair their functionality. Adhering to manufacturer guidelines for maintenance ensures that the detectors operate optimally at all times. Compliance and Regulations: Adhering to local building codes and fire safety regulations is essential. These codes often stipulate the number and placement of smoke detectors, ensuring comprehensive coverage throughout the building. Awareness and Education: Educating occupants about the importance of smoke detectors and how to respond when an alarm sound is crucial. Regular fire drills and safety education can enhance the effectiveness of the smoke detector system by ensuring that everyone knows what actions to take in an emergency. In summary, the implementation of a smoke detector system is a proactive and necessary step in safeguarding lives and property from the dangers of fire. By leveraging advanced technologies, ensuring regular maintenance, and adhering to safety regulations, these systems provide a robust defense against the potentially devastating effects of fire, thus creating safer living and working environments.Based on an experiment comparing different types of fire detectors, the conclusion is that ionization smoke detectors and rate of temperature rise heat detectors are the most sensitive to fires.

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